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METHODS FOR TREATING NEOPLASTIC DISEASES

Abstract

A method of treating a cancer, the method comprising: providing a modified macrophage or monocyte that contains an exogenous nucleic acid sequence encoding a Hom- 1 polypeptide or a fragment thereof that contains the Hom-1 homeobox domain, wherein the modified macrophage or monocyte expresses the Hom-1 polypeptide or the fragment thereof; and administering the modified macrophage or monocyte to a subject with a cancer.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of U.S. patent application Ser. No. 18/792,823, filed Aug. 2, 2024, which is a continuation of U.S. patent application Ser. No. 18/606,170, filed Mar. 15, 2024, now abandoned, which is a continuation of U.S. patent application Ser. No. 18/481,418, filed Oct. 5, 2023, now abandoned, which is a continuation of U.S. patent application Ser. No. 18/131,505, filed Apr. 6, 2023, now abandoned, which is a continuation of U.S. patent application Ser. No. 16/583,888, filed Sep. 26, 2019, now abandoned, which is a continuation of International Patent Application No. PCT/US2018/024764, filed on Mar. 28, 2018, which claims the benefit of the filing date of U.S. Provisional Application No. 62/477,754, filed on Mar. 28, 2017, and U.S. Provisional Application No. 62/516,401, filed on Jun. 7, 2017, the contents of each of which are hereby incorporated by reference herein in their entirety.

STATEMENT REGARDING SEQUENCE LISTING

[0002] The instant application contains a Sequence Listing XML file which has been submitted electronically in XML format and is hereby incorporated by reference in its entirety. Said XML copy, created on May 16, 2025, is named 130160-00107-SL.xml and is 6,355 bytes in size.

BACKGROUND OF THE INVENTION

[0003] Macrophages are executors of both innate and adaptive immunity, and have been recognized as a key component of tumors and their microenvironment. Extensive investigations suggested the role of tumor associated macrophages (TAMs) in the growth, invasion and metastasis of nearly all tumors. Derived from circulating monocytes, TAMs display a broad spectrum of phenotypes, ranging from the M1-like phenotype in early stages of selected tumors to the M2-like phenotype in most advanced tumors. Consistent with its role in promoting tumorigenesis, the M2-like TAMs display the characteristic phenotype with elevated expression of IL-10, IL4, MMP, VEGF, but decreased expression of pro-inflammatory cytokines and cytotoxic iNOs and ROIs implicated in tumoricidal activities. Besides their intrinsic function in promoting tumorigenesis, TAMs also contribute to the suppression of anti-tumor immunity by alternating T cells response and balance in tumor microenvironment. The functional plasticity of TAMs was well recognized. It has been proposed that by converting the pro-tumor M2-like TAMs into the anti-tumor M1-like phenotype, the TAMs could function as an attractive target for anti-tumor therapy. Over the past few years, multiple cell extrinsic and intrinsic factors have been explored for their potential function in converting TAMs into tumoricidal cells. However, the effectiveness and potential application of targeting TAMs in tumor treatment have been limited by the inadequate understanding of how the plasticity of TAMs is controlled by cell intrinsic factors.

[0004] Tumor-specific and tumor-associated antigens have been widely studied as targets for cancer immunotherapy. Targeting these antigens can potentially minimize off-target toxicities while increasing efficacy. On the other hand, antibodies against these antigens are often not effective on their own.

SUMMARY OF THE INVENTION

[0005] In one aspect, provided herein is a method of treating a cancer. The method comprises

providing a modified macrophage or monocyte that contains an exogenous nucleic acid sequence encoding a Hom-1 polypeptide or a fragment thereof that contains the Hom-1 homeobox domain, wherein the modified macrophage or monocyte expresses the Hom-1 polypeptide or the fragment thereof; and administering the modified macrophage or monocyte to a subject with a cancer. The modified macrophage exhibits an M1 phenotype. In some embodiments, the nucleic acid sequence is an mRNA molecule. In some embodiments, the exogenous nucleic acid sequence is operably linked to a heterologous or endogenous promoter. The method can further include administering an immune modulator to the subject.

[0006] In another aspect, described herein is a method of treating a cancer that includes contacting a macrophage or monocyte with one or more agents that induce expression of Hom-1, whereby the expression level of endogenous Hom-1 in the macrophage or monocyte is higher than before the contacting step; and administering the thus contacted macrophage or monocyte and an immune modulator to a subject with a cancer.

[0007] Also described herein is a method of treating a cancer that comprises contacting a macrophage or monocyte with an agent that induces the expression of an M1 gene or an agent that inhibits the expression of an M2 gene, whereby a macrophage that exhibits an M1-phenotype is generated; and administering the thus generated macrophage and an immune modulator to a subject with cancer.

[0008] In any of the methods, the immune modulator can be selected from the group consisting of CAR-T cells, immune check-point inhibitors, and antibodies against tumor-specific antigens, tumor-associated antigens, or neoantigens. In some embodiments, the neoantigen is CK20.

[0009] Any of methods disclosed herein can further include, prior to the administering step, detecting a lower level of Hom-1 expression in a tumor-associated macrophage in the subject as compared to a control.

[0010] The details of one or more embodiments are set forth in the description below. Other features, objects, and advantages of the embodiments will be apparent from the description and from the claims.

Description

DETAILED DESCRIPTION OF THE INVENTION

[0011] It was unexpectedly discovered that Hom-1 expression in TAMs is significantly decreased in comparison with macrophages isolated from normal tissues. It was further discovered that, also unexpectedly, increasing the expression of Hom-1 in TAMs converted them to M1-like macrophages with tumoricidal activities. In addition, it was shown that co-administration of Hom-1-expressing TAMs and an antibody against a neoantigen was surprisingly effective for suppressing tumor growth in vivo. As such Hom-1-regulated TAMs can be used as a new modality of cancer treatment, either alone, or in combinations with immune modulators such as CAR-T cells, immune check-point inhibitors, or antibodies against tumor-specific antigens, tumor-associated antigens or neoantigens, such as CK20 for colorectal cancers, ME1 for lung cancers, and CDC27 for melanomas.

[0012] Hom-1, a human homeobox transcriptional factor, is an antagonist of the canonical Wnt signaling. A nucleic acid sequence of Hom-1 (SEQ ID NO: 1) and the amino acid sequence (SEQ ID NO: 2) it encodes are disclosed herein. Positions 91-151 within SEQ ID NO: 2 encompass the homeobox domain.

[0013] Described herein is a method of treating a cancer in a subject by administering to the subject macrophages that exhibit anti-tumor activities. As used herein, unless otherwise specified, the terms “macrophages that exhibit anti-tumor activities,” “M1-like macrophages,” and “macrophages that exhibit an M1 phenotype” may be used interchangeably.

[0014] M1-like macrophages can be produced by (1) increasing Hom-1 expression in macrophages or monocytes, (2) increasing expression of one or more M1 genes in macrophages or monocytes, and/or (3) inhibiting expression of one or more M2 genes in macrophages or monocytes.

[0015] Monocytes (e.g., derived from a subject's peripheral blood) can be used in the method as it was shown that Hom-1 expression is both necessary and sufficient for monocyte-to-macrophage differentiation. In other words, increasing the expression of Hom-1 in monocytes can drive them to differentiate to macrophages.

[0016] Macrophages or monocytes can be induced ex vivo to express a higher level of endogenous Hom-1. Various agents or treatments can be used to induce Hom-1 expression, e.g., LPS, cholera toxin (CTX), chemotherapeutic agents, radiation, cytokines (e.g., GM-CSF), phorbol 12-myristate 13-acetate (PMA), and antibodies or RNAi against inhibitors of Hom-1 expression.

[0017] Modified macrophages or monocytes that exhibit an M1 phenotype can be generated by introducing into the macrophages or monocytes an exogenous mRNA molecule (e.g., a synthetic mRNA molecule) that contains a sequence encoding a Hom-1 polypeptide or a fragment thereof that includes the Hom-1 homeobox domain. Hom-1 expression in the macrophages or monocytes is thus increased transiently to induce an M1 phenotype. The modified macrophages or monocytes can then be administered to a cancer patient. The mRNA molecule can be a chemically-modified mRNA to promote stability of the mRNA and/or translation efficiency.

[0018] Macrophages or monocytes that have been genetically modified to express an elevated level of Hom-1 can also be used to treat a subject with cancer. For example, the genetically modified macrophages or monocytes can contain a nucleic acid sequence encoding a Hom-1 polypeptide or a fragment thereof that includes the Hom-1 homeobox domain. The nucleic acid sequence is operably linked to a promoter and the modified macrophages or monocytes express the Hom-1 polypeptide or the fragment thereof. Such modified macrophages or modified monocytes (which would differentiate to macrophages) express a sufficiently high level of Hom-1 to exhibit anti-tumor activities and/or an M1 phenotype.

[0019] Genetically modified macrophages or monocytes can also be generated by introducing extra copies of the Hom-1 gene into the macrophages or monocytes. For example, an expression construct containing a Hom-1 nucleic acid sequence (encoding a Hom-1 polypeptide or a fragment thereof that includes the Hom-1 homeobox domain) operably linked to the endogenous Hom-1 promoter can be introduced into macrophages or monocytes.

[0020] The expression level of endogenous Hom-1 in macrophages or monocytes can also be increased by genetically modifying regulatory elements of Hom-1 expression. For example, one or more negative transcription or translation regulatory elements of Hom-1 can be modified, deleted, or replaced to increase Hom-1 transcript and/or protein level. Genome editing techniques utilizing CRIPR, TALEN, or ZFNs or other techniques known in the art can be used to alter regulatory elements of Hom-1 expression.

[0021] A Hom-1 polypeptide or fragment thereof that includes the Hom-1 homeobox domain can also be introduced into macrophages or monocytes by direct peptide delivery.

[0022] Methods known in the art can be used to genetically modify macrophages and monocytes. For example, an exogenous expression construct for expressing Hom-1 can be introduced (e.g., stably or transiently transfected into) macrophages or monocytes. In one embodiment, the Hom-1 nucleic acid sequence is operably linked to a heterologous (i.e., not a Hom-1 promoter) constitutive or inducible promoter. In one embodiment, the Hom-1 nucleic acid sequence is operably linked to an endogenous promoter.

[0023] M1-like tumoricidal macrophages can also be generated by inducing expression of M1 genes in macrophages or monocytes. Agents that can induce M1 genes include, but not limited to, LPS, CTX, PMA, GM-SCF, INFO, and chemotherapeutic agents. Macrophages or monocytes can also be genetically modified to express elevated levels of M1 genes. M1 genes include IL1b, IL6, IL12, IL23, TNFa, iNOs, CD40, CD80, CD86, CD68, TLR4, TLR2, 25 IL-1R, MHCII, CCL15,

CCL20, CXCL9, CXCL1, and SOCS3.

[0024] Inhibiting expression of M2 genes in macrophages or monocytes can also produce M1-like tumoricidal macrophages. Agents that inhibit M2 genes include anti-IL4 agents (e.g., antibodies or RNAi agents), anti-IL13 agents (e.g., antibodies or RNAi agents), 30 antibodies against M2 proteins, and RNAi agents targeting M2 genes. M2 genes include ARG1, MMP9, CCL18, VEGF, IL10, IL4, TGFb, CD163, CD206, CD68., TLR8, TLR1, MHCII, TGM2, DcoyR, IL-1RII, Ym1/2, MMR/CD206, and SR.

[0025] Heterologous or autologous macrophages or monocytes can be used to generate M1-like macrophages. If heterologous macrophages or monocytes are used, HLA-matching can be conducted to avoid or minimize host reactions. HLA un-matched macrophages or monocytes may also be used. Autologous macrophages or monocytes can be obtained from a cancer patient using methods known in the art.

[0026] Immune modulators enhance, inhibit, or modulate one or more components of the immune system. Such modulators include CAR-T cells, immune check-point inhibitors, or antibodies against tumor-specific antigens, tumor-associated antigens or neoantigens, such as CK20 for colorectal cancers, ME1 for lung cancers, and CDC27 for melanomas.

[0027] The generated M1-like macrophages and/or an immune modulator can be administered to a subject through infusion or injection (for example, via intravenous, intrathecal, intramuscular, intraluminal, intratracheal, intraperitoneal, intracranial, subcutaneous, or another type of intra tissue route), transdermal administration, or other routes known in the art. In one example, the macrophages, monocytes, and/or immune modulator can be directly injected at a site or into a tissue (e.g., liver or pancreas) or its surrounding area, where a tumor is found.

[0028] The subject can be treated with the M1-like macrophages as often (e.g., every 1 to 30 days) and as many times (e.g., 1-30 times) as needed to treat the cancer. The M1-like macrophages described herein can also be used in a combination therapy with other cancer treatments such as radiation, chemotherapy, and small molecules drugs.

[0029] The data described below show that Hom-1 expression converts TAMs into tumoricidal cells independent of tumor types. Hence, any cancer can be treated using the M1-like macrophages described herein, particularly cancers associated with TAMs that express a low level of Hom-1. Before treating a subject with M1-like macrophages as described above, it may be useful to determine whether the Hom-1 expression level in TAMs in the subject is lower than that found in a control (e.g., a macrophage found in a normal tissue in the subject).

[0030] Examples of cancer that can be treated with M1-like macrophages include, but are not limited to, carcinoma and sarcoma such as leukemia, sarcoma, osteosarcoma, lymphomas, melanoma, glioma, glioblastoma, pheochromocytoma, hepatoma, ovarian cancer, skin cancer, testicular cancer, gastric cancer, pancreatic cancer, renal cancer, breast cancer, prostate cancer, colorectal cancer, cancer of head and neck, brain cancer, esophageal cancer, bladder cancer, adrenal cortical cancer, lung cancer, bronchus cancer, thyroid cancer, endometrial cancer, nasopharyngeal cancer, cervical cancer, liver cancer, metastatic cancer, and cancer of unknown primary site.

[0031] Detecting a lower expression level of Hom-1 or an M1 gene or a higher expression level of an M2 gene in macrophages found in the microenvironment of a tissue area as compared to that of a control (e.g., a corresponding level in a macrophage in a normal tissue) indicates that the tissue area is a cancer or is at risk of becoming a cancer. Thus, also contemplated herein is a method for identifying whether a suspected tissue area is a cancer or at risk of becoming a cancer.

[0032] Further described herein are methods of identifying a candidate compound for treating cancer. A test cell (i.e., a macrophage or monocyte) can be contacted with a test compound and the expression level of (i) Hom-1, (ii) a reporter gene operably linked to Hom-1 promoter, (iii) an M1 gene, (iv) a reporter gene operably linked to an M1 promoter, (v) an M2 gene, or (vi) a reporter gene operably linked to an M2 promoter in the test cell is detected. A test compound that increases the expression level of any of (i)-(iv), and/or decreases the expression level of (v) or (vi) as

compared to a control (e.g., a corresponding level in a test cell not contacted with the test compound) is a candidate compound for treating cancer.

[0033] In one screening method, a test compound is added to a co-culture containing a test cell and a cancer sample. A test compound is a candidate compound for treating cancer if it, as compared to a control, (i) increases the expression level of Hom-1, a reporter gene operably linked to a Hom-1 promoter, an M1 gene, or a reporter gene operably linked to an M1 promoter in the test cell, (ii) inhibits a significant decrease of the expression level of Hom-1, a reporter gene operably linked to a Hom-1 promoter, an M1 gene, or a reporter gene operably linked to an M1 promoter in the test cell, or (iii) decreases the expression level of an M2 gene or a reporter gene operably linked to an M2 promoter in the test cell.

[0034] In a co-culture system, whether a test compound has an inhibitory effect on the cancer sample can also be determined. A test compound that inhibits the cancer sample (e.g., inhibits growth of a cancer cell, kills a cancer cell, or decreases the size of the cancer sample) as compared to a control is a candidate compound for treating cancer. The test cell and the cancer sample can be in direct contact with each other. Alternatively, the test cell and the cancer sample are not in direct contact (e.g., with the use of transwell inserts). The cancer sample can be a sample containing a cancer cell, for example, a cancer tissue sample, a cancer cell isolated from a cancer tissue sample, or a cell of a cancer cell line. Cancer tissue samples can be obtained from surgically dissected specimens from cancer patients. Such cancer tissues samples may contain TAMs.

[0035] A screening method can also be performed with a cancer tissue sample in the absence of a test cell. A cancer tissue sample can be contacted with a test compound. After a time period, TAMs can be isolated from the cancer tissue sample and the expression level of Hom-1, an M1 gene, or an M2 gene in the TAMs can be determined. Alternatively or in addition to, the expression level of Hom-1 in the tissue sample can be determined. A test compound is a candidate compound for treating cancer if it, as compared to a control, (i) increases the expression level of Hom-1 or an M1 gene, (ii) inhibits a significant decrease of the expression level of Hom-1 or an M1 gene, and/or (iii) decreases the expression level of an M2 gene. A test compound that inhibits the cancer tissue sample (e.g., decreases the size of the sample) as compared to a control is also considered as a candidate compound for treating cancer.

[0036] Test compounds to be screened (e.g., proteins, peptides, peptidomimetics, peptoids, antibodies, RNAi, small molecules, or other drugs) can be obtained using a method known in the art.

[0037] In any of the methods described herein, the expression level of Hom-1, an M1 gene, or an M2 gene can be determined at either the mRNA level or at the protein level. Promoter activities can also be measured. Methods of measuring mRNA levels, protein levels, and promoter activities are well known in the art.

[0038] In any of the above-described screening methods, the test cell can be a macrophage or monocyte. The macrophage can be an M1 macrophage, an M2 macrophage, a tumor-associated macrophage, a tissue macrophage, or a monocyte-derived macrophage. The test cell can also be a monocyte.

[0039] The specific example below is to be construed as merely illustrative, and not limitative of the remainder of the disclosure in any way whatsoever. Without further elaboration, it is believed that one skilled in the art can, based on the description herein, utilize the present disclosure to its fullest extent. All publications cited herein are herein incorporated by reference in their entirety.

EXAMPLE

[0040] We identify a role of the human homeobox protein Hom-1 in functional polarization of human macrophages. We found that Hom-1 promotes and is required for M1 activation of human macrophages, but not required for the expression of critical genes involved in M2 activation.

[0041] Using primary TAMs isolated from cancers, we found that Hom-1 expression in TAMs was significantly decreased in comparison with macrophages isolated from normal control tissues. We

showed that the expression profile of Hom-1 in TAMs correlated with TAM phenotypes. Moreover, ectopic expression of Hom-1 converted TAMs into M1-like phenotype. Both in vitro and in vivo data showed that Hom-1 conferred tumoricidal activity to TAMs. Taken together, our studies demonstrated that Hom-1 converted TAMs into tumoricidal cells. We showed that Hom-1-expressing TAMs (exhibiting an M1 phenotype) exerted strong inhibitory effects on the growth of a variety of cancers, suggesting the role of Hom-1-modulated TAMs as a new modality in the treatment of cancers.

Hom-1 Expression was Decreased in TAMs

[0042] In advanced tumors, TAMs display a protumor M2-like phenotype. See, Bronte and Murray (2015), *Nat Med* 21, 117-119. The plasticity of TAMs has been well appreciated, and a variety of cytokines have been implicated in the polarization of TAMs towards M2 phenotype. See, Noy and Pollard (2014), *Immunity* 41, 49-61. In comparison, the transcriptional machinery that controls the TAM polarization remains largely unknown.

[0043] Discarded surgical specimens from colon cancer resections were used to isolate TAMs from tumor tissues as well as macrophages from normal mucosa 15 cm away from the tumor sites as described below. As previously noted, FACS analysis showed that, in comparison to macrophages isolated from normal control mucosa, TAMs expressed significant higher levels of cell surface markers associated with M2 phenotypes, such as the CD68, CD163, CD206. See, Zhang et al. (2013), *Eur J Cancer* 49, 3320-3334. We found that there was no significant difference in the expression of non-discriminating macrophage marker CD33 in TAMs and control macrophages.

[0044] To determine whether Hom-1 may play a role in TAMs, we quantified Hom-1 expression in TAMs by qRT-PCR and found that, in comparison with its expression in control macrophages, Hom-1 expression is significantly decreased in TAMs. VentX regulated TAM plasticity and polarized TAMs toward an M1 phenotype

[0045] Previous studies indicated that TAMs can be induced by LPS to display an M1 phenotype. See, Zhang et al. To determine whether Hom-1 plays a role in TAM plasticity, we examined Hom-1 expression in TAMs exposed to LPS. We found that Hom-1 expression was significantly elevated in TAMs after being stimulated with LPS. Parallel with the elevated expression of Hom-1 and consistent with prior findings, LPS stimulation of TAMs led to elevated secretion of inflammatory cytokines and cytotoxic iNOs.

[0046] To determine whether Hom-1 plays a regulatory role of TAM plasticity, we examined the effects of Hom-1 knockdown on TAM phenotype. Treatment of TAMs with anti-Hom-1 morpholinos (MO) led to around an 80% reduction of Hom-1 expression. Consistent with Hom-1's role as a key regulatory of TAM plasticity, we found that Hom-1 MO aborted LPS induced secretion of inflammatory cytokines and cytotoxic iNOs in TAMs. CD206 is a mannose receptor and a M2 cell surface marker that is highly expressed in TAMs. Reflecting the plasticity of TAMs, exposure of TAMs to LPS led to a significant reduction of CD68+CD206+ population and an increase of the CD68+CD206- cell numbers. Hom-1 MO abolished both effects of LPS on TAMs ($p < 0.01$).

[0047] The correlation between Hom-1 expression levels and TAM phenotypes prompted us to further explore the idea that Hom-1 controls TAM plasticity. TAMs were isolated and transfected with a plasmid encoding GFP-Hom-1 or control GFP. Compared with the control GFP transfected TAMs, TAMs transfected with GFP-Hom-1 displayed characteristics M1 morphology with elongated/fibroblast-like cell shape. FACS analysis showed that surface expressions of M1 markers CD40, CD80, and CD86 were significantly increased in TAMs transfected with GFP-Hom-1. In addition, secretions of pro-inflammatory cytokines TNF α , IL-1 β , and IL-12 were significantly increased, while secretion of M2 cytokine IL-10 was significantly decreased in TAMs transfected with GFP-Hom-1. Consistently, gene expression analysis showed that M1 genes, such as IL-1 β , IL-6, TNF- α , and iNOs increased significantly in TAMs transfected with GFP-Hom-1, while M2 genes, such as, CCL18, MMP9, VEGFA, and Arg1 decreased significantly in TAMs transfected

with GFP-Hom-1. Taken together, our data suggested that Hom-1 regulated TAM plasticity and promoted M1 polarization of TAMs.

Hom-1 Converted TAMs Into Tumor Suppressing Cells

[0048] Our findings that Hom-1 promotes M1 polarization of TAMs prompted us to explore whether Hom-1-modified TAMs could exert tumor suppression. Freshly isolated TAMs from colon cancers were transfected with plasmid encoding GFP-Hom-1 or control GFP. The modified TAMs were then co-cultured with tumor or normal tissues of the same patient, using the trans-well culture system. Remarkably, after 7-10 days of co-culturing, tumor volumes were significantly decreased (around 70%) during the incubating with GFP-Hom-1 modified TAMs ($p < 0.01$), while there was no significant change of size in tumors incubated with GFP-transfected TAMs or with TAMs alone. To determine whether the shrinkage of tumor volume was related to reduction of cancer cells, we performed tissue section, H&E staining and immunohistochemistry of colon cancer cells with CK20 antibody. We found that CK20 positive tumor cells appeared in nests, cords, and sheets in the tumors incubated with TAMs or GFP modified TAM. However, CK20 positive tumor cells disappeared during the incubation with TAMs transfected with GFP-Hom-1. The specificity of tumoricidal effects of Hom-1-modified TAMs was demonstrated by the findings that Hom-1-modified TAMs exerted minimal effects on the volume and morphology of normal colon mucosa during the incubation with TAMs transfected with either GFP or GFP-Hom-1.

Hom-1 Promotes Tumoricidal Function of TAMs in Vivo

[0049] Our findings that Hom-1 converted TAMs into tumoricidal cells in vitro prompted us to explore the potential role of Hom-1-regulated TAMs in tumorigenesis in vivo. Colon cancers were cut into around 0.5 cm pieces and surgically seeded in the subcutaneous space of the abdominal side of NSG mice. One week later, MO-Hom-1 transfected TAMs (Hom-1 inhibited) or GFP-Hom-1 transfected TAMs (Hom-1 expressing) were injected through mouse tail vein. After 8 weeks of xenograft, tumors developed in mice injected with MO-Hom-1 transfected TAMs, but not in mice injected with GFP-Hom-1 transfected TAMs.

[0050] In addition, we evaluated the effect of an anti-CK20 antibody in combination with MO-Hom-1 transfected TAMs in the same mouse model. The antibody alone or in combination with MO-Hom-1 transfected TAMs were administered to the mice after tumors have formed. The tumors in mice administered with the antibody alone continued to grow. On the other hand, the tumors in mice administered with the antibody and MO-Hom-1 transfected TAMs ceased to grow or grew at a much slower rate in comparison to the mice treated with the antibody alone.

[0051] We also found that the TAMs or monocytes can be induced to exhibit an M1 phenotype by culturing them in an M1 differentiation media. These M1-differentiated TAMs/monocytes can be infused into NSG mice and inhibit cancer growth in vivo. The effects of M1 differentiated TAMs on tumor growth are abolished by inhibition of Hom-1 expression in these TAMs or monocytes.

Effects of Ectopic Expression of Hom-1 in TAMs on Various Cancer Types

[0052] TAMs have been implicated in oncogenesis of essentially all tumors. Following our studies on TAMs in colon cancer cells, we extended our investigation to other tumor types.

[0053] Surgical species of lung, melanoma, esophagus, gastric, and pancreatic cancers were obtained, and TAMs were isolated as described above. Macrophages from corresponding normal tissues of the same patient were obtained. Hom-1 expression in TAMs and tissue macrophages were quantified using real time RT-PCR. Hom-1 expressions in TAMs of all these tumors were low in comparison to Hom-1 expression in corresponding macrophages from distant normal tissues.

[0054] To determine whether Hom-1 could convert these TAMs into tumoricidal cells, GFP or GFP-Hom-1 were transfected into the TAMs. After 48 hours of transfection, GFP positive cells were sorted out and co-cultured with individual tumors. Tumor volumes of all tumors decreased during co-culture with GFP-Hom-1 transfected TAMs but not with the control GFP-transfected TAMs. Our results suggested that Hom-1 could convert TAMs into tumoricidal cells independent of tumor types.

Collection of Colon Tissue Samples

[0055] Cancer tissues and normal tissues were obtained from surgically dissected specimens from patients in pathology lab. Around 5-10 gram of tissue was collected from each tumor mass, or from normal mucosa at 15 cm away from tumor mass. Patient blood samples were also collected.

Preparation of Intraepithelial Lymphocyte

[0056] Lamina propria mononuclear cells (LPMCs) were isolated using previously described techniques with modification (Kamada N, et al, 2008; Pignata C, et al, 1990). In brief, dissected fresh mucosa and tumor mass were rinsed in 10-cm Petri dish with Ca.sup.2+-free and Mg.sup.2+-free hank's balanced salt solution (HBSS) (life technologies) containing 2% fetal bovine serum (FBS) and 1 mM Dithiothreitol (DTT) (Sigma-Aldrich) to remove mucus. The mucosa and tumor were cut into 0.5 cm pieces by a razor blade and incubated in 6-well plate with 5 mL HBSS containing 1 mM EDTA (Sigma-Aldrich) at 37° C. for 1 hour, then passed through a gray-mesh (100 micron). The flowthrough contains intraepithelial lymphocyte and epithelial cells and was analysis by a flow cytometer.

Isolation of Macrophage From Tumor Mass and Normal Mucosa

[0057] Subsequently, the mucosa and tumor were incubated in HBSS (with Ca.sup.2+ and Mg.sup.2+), containing 2% FBS, 1.5 mg/mL Collagenase D (Roche), 0.1 mg/mL Dnase I at 37° C. for 1 hour. Digested tissues were passed through a gray-mesh (70 micron) filter. The flowthrough was collected and resuspended in a 40% Percoll solution (Pharmacia), then layered on 60% Percoll, and centrifuged at 2000 rpm for 30 min without brake. LPMCs at the interface were collected. Normal mucosal macrophages and TAMs were purified from LPMCs using EasySep™ Human Monocyte/Macrophage Enrichment kit without CD16 depletion (StemCell Technologies), according to the manufacturer's instructions. Cells isolated by these techniques were routinely more than 98% viable by propidium iodide (PI) staining. The purity of intestinal macrophages was more than 95%.

Preparation of Macrophages From Peripheral Blood

[0058] Peripheral blood mononuclear cells (PBMC) from healthy adult donors at Brigham and Women hospital were isolated by Ficoll density gradient centrifugation. Human monocytes were purified from PBMCs using EasySep™ Human Monocyte Enrichment kit without CD16 depletion according to the manufacturer's instructions. Purified cells were cultured in completed RPMI medium with 10 ng/mL of M-CSF (PeproTech). After enrichment, monocytes were cultured in completed RPMI medium with M-CSF for 5 days, cells were used for co-culture system.

FACS Analysis

[0059] Phenotypic analysis of TAMs and other lymphocytes was performed using flow cytometry after immunolabeling of cells with fluorescence dye-conjugated antibodies. The following antibodies were used: PE-conjugated anti-CD3 (OKT3), -CD25 (BC96), -CD14 (61D3), -CD68 (eBio Y182A), -CD163 (eBio GH161), -CD206, FITC-conjugated anti-CD4 (RPA-T4), -CD33 (HIM3-4), APC-conjugated anti-CD8 (OKT8), -CD4 (OKT4) (eBioscience, Inc). Intracellular staining of Foxp3 (236A/E7), IFN- γ , Perforin, and Granzyme B was performed with PE-conjugated antibodies following the protocol provided by manufacturer. Isotope control labeling was performed in parallel. Antibodies were diluted as recommended by the supplier. Labeled cells were collected on FACScan flow cytometer with Cell-Quest software (BD Biosciences) and analyzed by FlowJo software. Results are expressed as the percentage of positive cells.

Organotypic Co-Cultures of Tumor and Macrophages

[0060] Transwell inserts (0.4 μ m pore sized, Costar, Corning) were placed in 12-well polystyrene tissue culture plates (Becton Dickinson, Franklin Lakes, NJ). Mucosa and tumor mass were weighed and washed with 1x PBS buffer, plus antibiotics, then cut into 0.5 cm pieces. Around 50 mg of tissues were seeded in the upper compartment of a 12-well transwell and filled with 0.5 mL of RPMI 1640 completed medium. 5×10^5 of TAMs were added to the lower compartment at the density of 0.5 million cells/well with no direct cell-tissue contact and filled with 2 mL of PRMI

completed medium. The plate incubated at 37° C., 5% CO₂. 0.5 mL of culture medium were collected for cytokine analysis and fresh medium were added every three days. After two weeks co-culture, low chamber macrophages were collected and total RNA was isolated by the TRIzol reagent (Ambion). Tumor and normal mucosal were monitored two times a week by calipers to calculate tumor volumes according to the formula (length×width²)/2. The photos of tissues were taken by a 3-megapixel CMOS camera on Leica EZ4D stereomicroscope or a digital camera on iPhone.

Immunohistochemistry

[0061] Tumors or normal tissues were fixated in Formalin (Fisher Scientific Company, Kalamazoo, MI). CK20 stainings (Dako, Carpinteria, CA, clone Ks20.8, 1:50) and Haematoxylin/eosin (H&E) stainings were performed. CK20 staining was performed on the Leica Bone III staining platform using Epitope Retrieval 2 for 20 minutes online, and using Bone Polymer Refine detection kit. Microscopic analysis was performed with a Nikon Eclipse Ti fluorescence microscopy. Images were captured at an original magnification of 40× using a color camera applying the NIS Elements imaging software (Nikon). Brightness and contrast for representative images were adjusted equally among groups.

Hom-1 Overexpression

[0062] Transfection of GFP-Hom-1 into blood macrophages and TAMs was carried out through lipofectamine 2000 (Life technologies) according to manufacture protocol. 48 hours after transfection, cells were filtered through a 70 um filter for cell sorting. GFP positive cells were sorted by BD FACSAria II under the Baker Bio-Protect Hood in a sterile condition. After sorting, cells were cultured in RPMI 1640 completely medium.

Hom-1 knockdown

[0063] Colon TAMs or human primary monocytes were transfected with Morpholino (MO) antisense oligonucleotides using the Human Monocyte Nucleofector Kit (Lonza, Walkersville, MD) according to the manufacturer's instructions. Briefly, 5×10⁶ cells were re-suspended into 100 µl nucleofector solution with 2.5 nmol of either Hom-1 MO oligonucleotides or a standard control MO oligonucleotides and electroporated with the Nucleofector II Device (Lonza). Cells were then immediately removed from the device and incubated overnight with 1ml pre-warmed Human Monocyte Nucleofector Medium containing 2 mM glutamine and 10% FBS. Cells were then re-suspended into complete RPMI medium and treated with appropriate cytokines to induce differentiation into macrophages. All the MO oligonucleotides were ordered from Gene Tools (Philomath, OR).

Cytokine measurement

[0064] Levels of IL-1β, IL-10, TNF-α and IL-12p70 in the supernatants of E. coli LPS-(Sigma-Aldrich) treated blood macrophages or LPS-treated TAMs were quantified using ELISA kits obtained from eBiosciences. Analyses were conducted according to the manufacturer's instructions. qRT-PCR

[0065] Total RNA was isolated by the TRIzol reagent and RNA amounts were measured by NanoDrop 2000 (Thermo Scientific). An equal amount of RNA was used for first-strand cDNA synthesis with SuperScript III First-Strand Synthesis System (Life Technologies) according to the manufacturer's protocol. To amplify Hom-1 cDNA with conventional PCR, we used the AccuPrime Taq DNA polymerase system (Life Technologies) following the manufacturer's instructions. PCR products were separated on 2% agarose gels and stained with ethidium bromide. GAPDH was used as an internal control. We performed quantitative measurement of Hom-1 and other genes cDNA with SYBR Green on a LightCycler (480 Real-Time PCR System; Roche). Relative expression profiles of mRNAs were then calculated using the comparative Ct method (AACt method).

Arginase Activity and NO Assays

[0066] Arginase activity was quantified in cell lysates by measuring the production of urea using the QuantiChrom arginase Assay Kit (DARG-200; BioAssays Systems). Nitrite concentrations in

culture supernatants were determined using Griess reagent kit (Molecular Probes).

Statistical Analysis

[0067] Student's test was used for statistical analysis.

OTHER EMBODIMENTS

[0068] All of the features disclosed in this specification may be combined in any combination. Each feature disclosed in this specification may be replaced by an alternative feature serving the same, equivalent, or similar purpose. Thus, unless expressly stated otherwise, each feature disclosed is only an example of a generic series of equivalent or similar features.

[0069] From the above description, one skilled in the art can easily ascertain the essential characteristics of the described embodiments, and without departing from the spirit and scope thereof, can make various changes and modifications of the embodiments to adapt it to various usages and conditions. Thus, other embodiments are also within the claims.

Claims

1. A method of treating a cancer, the method comprising: providing a modified macrophage or monocyte that contains an exogenous nucleic acid sequence encoding a Hom-1 polypeptide or a fragment thereof that contains the Hom-1 homeobox domain, wherein the modified macrophage or monocyte expresses the Hom-1 polypeptide or the fragment thereof; and administering the modified macrophage or monocyte to a subject with a cancer.
2. The method of claim 2, wherein the modified macrophage or monocyte is generated by introducing the exogenous nucleic acid into a macrophage or monocyte derived from the subject or another subject.
3. The method of claim 1, wherein the exogenous nucleic acid is an mRNA molecule.
4. The method of claim 3, wherein the mRNA molecules is a chemically modified mRNA molecule.
5. The method of claim 1, wherein the modified macrophage exhibits an M1 phenotype.
6. The method of claim 1, wherein the exogenous nucleic acid sequence is operably linked to a heterologous or endogenous promoter.
7. The method of claim 6, wherein the promoter is a constitutive promoter or an inducible promoter.
8. (canceled)
9. The method of claim 1, further comprising administering an immune modulator to the subject.
10. The method of claim 9, wherein the immune modulator is selected from the group consisting of CAR-T cells, immune check-point inhibitors, and antibodies against tumor-specific antigens, tumor-associated antigens, or neoantigens.
11. The method of claim 10, wherein the neoantigen is CK20.
12. The method of claim 1, further comprising, prior to the administering step, detecting a lower level of Hom-1 expression in a tumor-associated macrophage in the subject as compared to a control.
13. The method of claim 1, wherein the cancer is selected from the group consisting of leukemia, sarcoma, osteosarcoma, lymphomas, melanoma, glioma, glioblastoma, pheochromocytoma, hepatoma, ovarian cancer, skin cancer, testicular cancer, gastric cancer, pancreatic cancer, renal cancer, breast cancer, prostate cancer, colorectal cancer, cancer of head and neck, brain cancer, esophageal cancer, bladder cancer, adrenal cortical cancer, lung cancer, bronchus cancer, thyroid cancer, endometrial cancer, nasopharyngeal cancer, cervical cancer, liver cancer, metastatic cancer, and cancer of unknown primary site.
14. A method of treating a cancer, the method comprising: contacting a macrophage or monocyte with one or more agents that induce expression of Hom-1, whereby the expression level of endogenous Hom-1 in the macrophage or monocyte is higher than before the contacting step; and

administering the thus contacted macrophage or monocyte and an immune modulator to a subject with a cancer.

15. The method of claim 14, wherein the immune modulator is selected from the group consisting of CAR-T cells, immune check-point inhibitors, and antibodies against tumor-specific antigens, tumor-associated antigens, or neoantigens.

16. The method of claim 15, wherein the neoantigen is CK20.

17. The method of claim 14, wherein the macrophage or monocyte is autologous or heterologous to the subject.

18. The method of claim 14, wherein the contacting step and the administering step are repeated at least once.

19. A method of treating a cancer, the method comprising: contacting a macrophage or monocyte with an agent that induces the expression of an M1 gene or an agent that inhibits the expression of an M2 gene, whereby a macrophage that exhibits an M1-phenotype is generated; and administering the thus generated macrophage and an immune modulator to a subject with cancer.

20. The method of claim 19, wherein the macrophage or monocyte is autologous or heterologous to the subject.

21. The method of claim 19, wherein the immune modulator is selected from the group consisting of CAR-T cells, immune check-point inhibitors, and antibodies against tumor-specific antigens, tumor-associated antigens, or neoantigens.

22. (canceled)
